

Name:

Key

SKILLS QUIZ

1. a) In the following matrix-vector multiplication over the real numbers, find the values of a and b .

$$\begin{pmatrix} 2 & 1 & 0 \\ -2 & 1 & 4 \\ b & 0 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ a \\ -3 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 & 0 \\ -2 & 1 & 4 \\ b & 0 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 14 \\ b-8 \end{pmatrix}$$

so

$$\text{and } b-8 = -3 \Rightarrow$$

$$a = 14$$

$$b = 5$$

- b) Give an example of a matrix that represents a linear transformation using the standard basis such that its domain is $\mathbb{R}^2$ and its codomain is $\mathbb{R}^4$, or explain why this is not possible.

Any 4×2 real matrix is a correct answer:

eg.

$$\begin{pmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 0 \\ 4 & 8 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 0 \\ 4 & 8 \end{pmatrix} \begin{pmatrix} - \\ - \end{pmatrix} = \begin{pmatrix} - \\ - \\ - \\ - \end{pmatrix}$$

domain is $\mathbb{R}^2$ codomain is $\mathbb{R}^4$

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2. a) For each of the following matrix operations, calculate the result if the operation is defined. Otherwise, state that the operation is undefined.

$$A = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 1 & -1 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 2 & 0 & -6 \\ 0 & 1 & 2 \end{pmatrix} \quad D = \begin{pmatrix} 0 & 0 \\ 3 & -2 \\ 0 & 1 \end{pmatrix}$$

$A + C$, AB , DC , BA

$$A + C = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 1 & -1 \end{pmatrix} + \begin{pmatrix} 2 & 0 & -6 \\ 0 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 0 & -3 \\ 0 & 2 & 1 \end{pmatrix}$$

$$AB = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} \leftarrow \text{undefined}$$

$$DC = \begin{pmatrix} 0 & 0 \\ 3 & -2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & -6 \\ 0 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 6 & -2 & -22 \\ 0 & 1 & 2 \end{pmatrix}$$

$$BA = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 3 \\ 0 & 1 & -1 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 6 \\ 2 & 1 & 2 \end{pmatrix}$$

b) Give an example of two matrices whose product is the matrix M ; or explain why this is not possible.

$$M = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}$$

There are many possible answers. For instance,

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = M$$

$$\text{also } \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 19 & 25 & \pi \end{pmatrix} = M$$

$$\text{also } \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$